

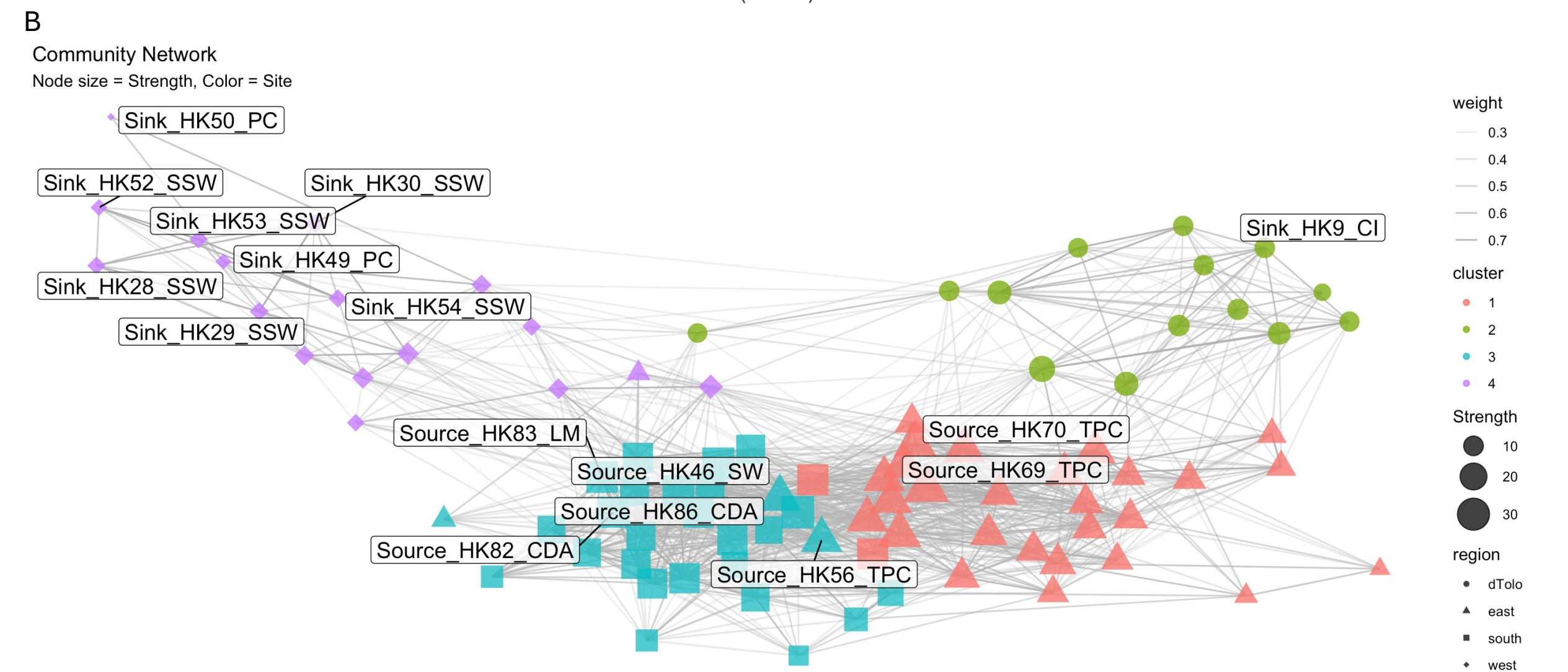
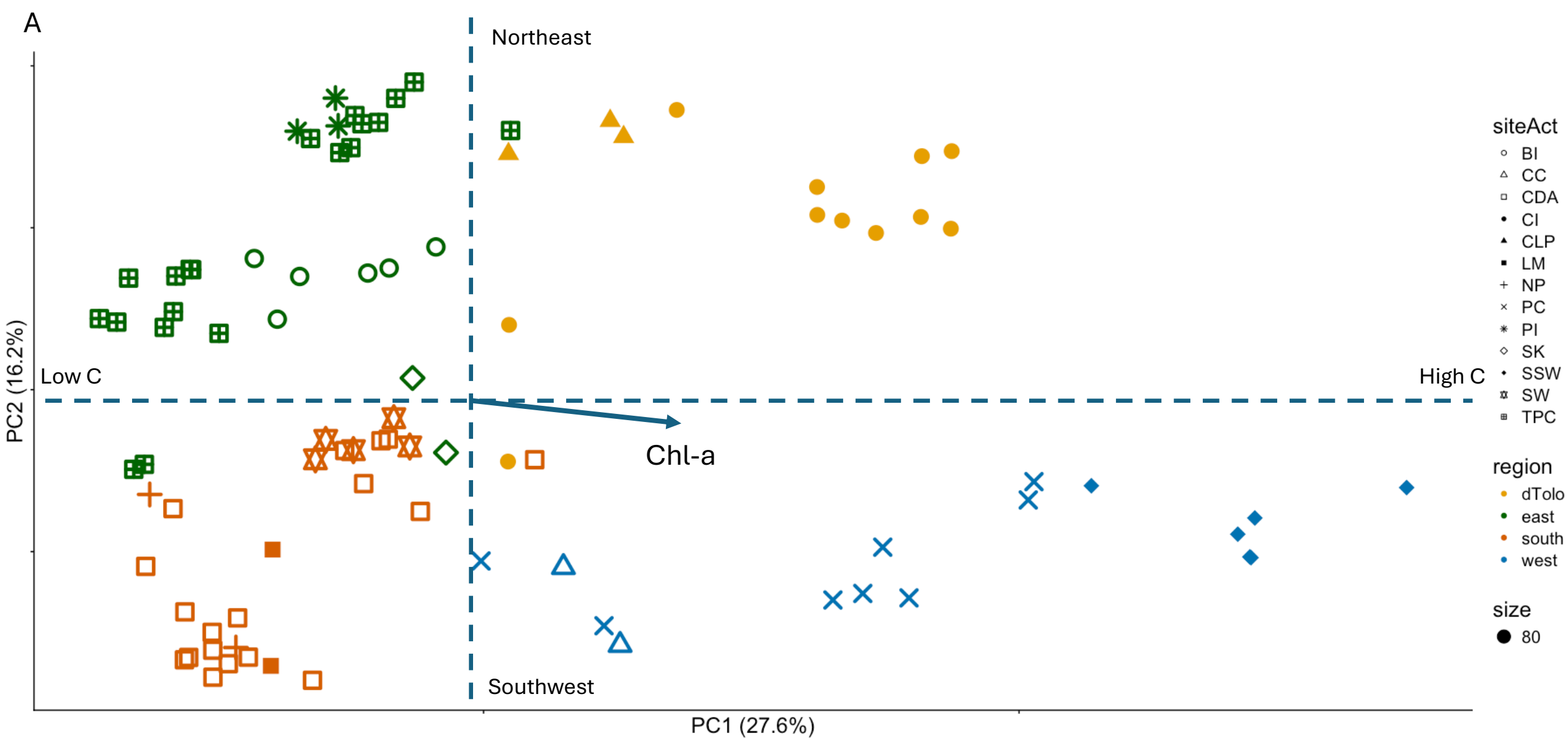


Figure 3: PCoA based on Bray-Curtis dissimilarity matrix (A). Communities from all four regions (different color) were significantly different from each others. Chlorophyll-*a* concentration significantly correlated with primary directional shift. Multi-level modularity optimization (B) showed that metacommunity modularity aligned well regionally as shapes and colors were largely (93.2%) in concurrence. The source communities (top 10% in source index) and the sink communities (bottom 10% in source index) were annotated.